20 Donna found out information about the areas, in m^2, of some fields.

The table shows some information about her results.

Area (p m^2)	Frequency
$0 < p \leqslant 10\,000$	4
$10\,000 < p \leqslant 30\,000$	16
$30\,000 < p \leqslant 60\,000$	15
$60\,000 < p \leqslant 70\,000$	
$70\,000 < p \leqslant 110\,000$	

Donna drew a histogram of her results.

The height of the bar for the class interval $70\,000 < p \leqslant 110\,000$ is twice the height of the bar for the class interval $0 < p \leqslant 10\,000$

(a) Write down the frequency for the interval $70\,000 < p \leqslant 110\,000$

Last bar: twice the density and four times the width.

So its frequency = 2 × 4 × 4 = 32.

1 mark
Area of bar represents frequency.

.....

(1)

The height of the bar for the class interval $10\,000 < p \leqslant 30\,000$ is 4 cm.

(b) Work out the height of the bar for the class interval $30\,000 < p \leqslant 60\,000$

Known density = 16 ÷ 20000 = 0.0008

1 mark
Frequency ÷ class width;
height proportional to density.

Required density = 15 ÷ 30000 = 0.0005

Height = 4 × 0.0005 ÷ 0.0008 = 2.5 cm

1 mark
2.5 cm.

..... cm

(2)

(Total for Question 20 is 3 marks)

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